Five Independent Dimensions of Game Design Space: Evidence from Principal Component Analysis

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Abstract

Game design space has been measured almost exclusively on a single quality axis. We apply principal component analysis to the TIGRIS corpus (150 games, 32 axes) and show that normalization choice is decisive: raw PCA yields PC1 = 81.3%, confirming a dominant general quality factor; weight normalization reduces this to 72.6%; within-game z-score normalization — centering each game on its own axis mean — reduces PC1 to 50.0% and requires 4 components to explain ≥80% of variance. The resulting components correspond to five theoretically coherent and empirically separable design dimensions: Elegance/Entry (50%), Game-Architecture (18.2%), Tension profile (9.9%), Interaction geometry (6.3%), and Range/ergodicity (residual). These five dimensions — named TIGER in the companion framework paper TG-B1 — are confirmed orthogonal by 42 independent collision votes from the adversarial Parliament review procedure applied across the corpus. The within-game normalization also reveals a genuine design trade-off: Information Architecture Type (A32) negatively correlates with Count-Robustness ($r = -0.779$), confirming that hidden-identity and inverted-visibility games belong to a structurally distinct design paradigm.

1 Introduction

How many independent ways can board games differ from one another? This question is not merely taxonomic — it is foundational for any framework that claims to characterize game design rather than rank it. If most variation in multi-axis scores is explained by a single quality factor, then dimensional frameworks are window-dressing on a single underlying continuum. If multiple genuinely independent dimensions exist, then games occupy a space, and that space can be navigated.

1.1 The General Factor Problem

Raw principal component analysis of multi-axis game scores consistently produces a dominant first component that we call the *general quality factor*: all axes load positively on PC1, and PC1 explains the majority of total variance [3]. This is expected — better-designed games tend to score higher on all axes simultaneously, producing positive intercorrelations that PCA collapses into a single dimension. The general factor is real, but it is not useful for characterizing how games differ.

1.2 The Normalization Insight

The central methodological contribution of this paper is the demonstration that *within-game z-score normalization* — computing each axis score relative to that game's own mean and standard deviation across all axes — suppresses the general factor and reveals the structure of residual variation. The result is a space of design distinctiveness rather than design quality, and PCA on this normalized space yields 4–5 genuinely independent components that correspond to theoretically coherent design dimensions.

2 Related Work

Principal component analysis has been applied to design evaluation in several domains. In industrial design, Yanagisawa and Fukuda [5] used PCA to identify independent aesthetic dimensions from multi-attribute product ratings. In music, Aucouturier and Pachet [1] applied PCA to timbral features to identify orthogonal perceptual dimensions. In game studies, quantitative approaches have been limited by corpus size: most studies work with fewer than 50 games and rely on community ratings rather than structured axis scores.

The general factor problem we address — where PCA on correlated evaluative ratings yields a dominant quality PC — is well-known in psychometrics as the *g factor* problem [4] and in consumer research as the halo effect. Within-group normalization to suppress the general factor has been used in evolutionary biology [2] and personality psychology but has not, to our knowledge, been applied to game design analysis.

3 Methods

3.1 Data

We use the TIGRIS corpus of 150 games scored on 32 axes [3]. All axes are normalized to a 1–10 scale before analysis. Games with missing values on more than 4 axes ($< 2\%$ of the corpus) are excluded from PCA analysis.

3.2 Three Normalization Approaches

We compare three normalization methods applied before PCA:

Raw (N1): No normalization beyond per-axis z-scoring across all games. This is the standard approach and produces the dominant general quality PC.

Weight-normalized (N2): Each axis score is divided by BGG weight before z-scoring, removing the portion of variance explained by overall game complexity. This partially suppresses the general factor.

Within-game z-score (N3): For each game, we compute the mean and standard deviation of that game's scores across all 32 axes and z-score each axis score relative to that game's own distribution. This removes the game-level quality signal entirely, leaving only the pattern of relative axis emphasis.

3.3 PCA and Scree Analysis

PCA is applied to each normalized dataset using singular value decomposition. We report: scree plots showing variance explained per component, component loadings for the top 5 PCs, and the top 3 axis correlates of each PC under N3. Component labels are assigned post-hoc based on the highest-loading axes.

4 Results

4.1 Normalization Comparison

Table 1 reports the variance explained by the top 5 principal components under each normalization method. The effect of normalization choice is decisive.

Table 1: Variance explained by top 5 PCs under three normalization methods (150 games, 32 axes).

Normalization	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**Cum.**
N1 (Raw scores)	81.3%	3.9%	2.7%	2.3%	2.1%	92.3%
N2 (Weight-normalized)	72.6%	8.1%	5.9%	3.3%	2.8%	92.7%
N3 (Within-game z-score)	50.0%	18.2%	9.9%	6.3%	5.1%	89.5%

Components for 80% variance: N1=1, N2=2, N3=4.

Under N1, a single dominant component explains 81.3% of variance, reflecting the general quality factor (games that score high on one axis tend to score high on all). N2 reduces this to 72.6% by removing the correlation with BGG weight, revealing PC2 as a production-architecture factor. N3 further reduces PC1 to 50.0%, requires 4 components for 80% coverage, and distributes variance more evenly across components.

4.2 Top Axis Correlations and the General Factor

Under N3 normalization, the strongest correlations reveal the structure of the residual PC1 (50%): the "accessibility bundle" (A1 Elegance, A6 Teachability, C7 Action-Menu Clarity, D3 Count-Robustness, A5 Pacing) co-vary at $r > 0.95$ in within-game normalized space. These axes measure *the same underlying construct*: how efficiently the game creates engagement relative to its own average.

Critically, the normalization also reveals genuine *negative* correlations: A32 (Information Architecture Type) negatively correlates with D3 ($r = -0.779$), D1 ($r = -0.777$), and A2 ($r = -0.777$). Games with inverted-visibility or hidden-identity information structures are *incompatible* with high count-robustness and decision-density — this is a real design trade-off, not a measurement artifact.

4.3 Component Loadings Under N3

Table 2 shows the top axis loadings for each PC under N3.

Table 2: Top axis loadings per PC under N3 (within-game z-score normalization).

PC	% var.	Top positive loadings	Top negative loadings
PC1	50.0	A2(+0.25), C1(+0.20), A7(+0.20), C7(+0.20)	A31(-0.26), A32(-0.25)
PC2	18.2	C3(+0.33), D4(+0.33), B3(+0.33), B1(+0.22)	A26(-0.07)
PC3	9.9	B2(+0.39), C1(+0.27), A3(+0.24)	B6(-0.32), A5(-0.28)
PC4	6.3	A28(+0.44), B4(+0.39), A3(+0.24)	D2(-0.44), A32(-0.37)
PC5	5.1	C8(+0.58), A25(+0.37)	D2(-0.46), A5(-0.22)

4.4 Naming the Five Dimensions

The four components required for 80% coverage, plus the structured PC1, correspond to five theoretically coherent design dimensions. Following the TIGER acronym established in TG-B1, we assign labels *by construct*, not by PCA order:

E — Elegance/Entry corresponds to PC1's accessibility bundle (A1, A6, C7, D3, A5 co-vary strongly). Measures: how efficiently rules create engagement. *Canonical high*: Crokinole, Codenames, Azul.

G — Game-Architecture corresponds to PC2 (18.2%). Measures: depth of the production/resource pipeline. C3 (scarcity), D4 (lock-in), B3 (conversion chain) define this component. *Canonical high*: Brass: Birmingham, Agricola, HAUL.

T — Tension corresponds to PC3 (9.9%). B2 (catastrophe pressure) loads strongly positive; A5 (pacing) loads negative — punishing games move slower. Measures: what kind of pressure the game creates. *Canonical high*: CRUCIBLE, Spirit Island, Agricola.

I — Interaction corresponds to PC4 (6.3%). D2 (spatial interaction) loads negative; A28 (cognitive load type) and B4 (information cost) load positive. Separates *spatial contest* games (Crokinole, Chess) from *informational contest* games (Hanabi, Secret Hitler).

R — Range corresponds to PC5 and residual variance around A30 (Strategy Ergodicity). Measures: how many viable strategic paths exist. *Canonical high*: Dominion, Wingspan. *Canonical low*: Chess.

Note that component order differs from TIGER naming order. E (Elegance) corresponds to PC1 (largest), G (Architecture) to PC2, T (Tension) to PC3, I (Interaction) to PC4, and R (Range) to residual PC5. The TIGER acronym is ordered for memorability, not by variance contribution.

5 Discussion

The central finding — that within-game z-score normalization reveals 4–5 genuinely independent dimensions where raw PCA shows only one — has a clear practical implication: the choice of normalization method is not a technical detail but a theoretical commitment. Raw PCA answers the question "how much of the variation between games is explained by overall quality?" (answer: most of it). Within-game normalization answers the different question "how much of the variation in design character is explained by orthogonal design dimensions?" (answer: the five TIGER dimensions explain $\geq 80\%$).

5.1 What Each Dimension Represents

Tension (T). The axes that load on this dimension — time pressure, catch-up difficulty, penalty severity, and elimination risk — collectively encode the psychological pressure experienced by the player. This maps to Knizia's oft-cited design principle that meaningful decisions require real stakes: without Tension, choices are preferences rather than decisions. The empirical orthogonality of Tension from Interaction confirms that pressure can be generated by the game system (solo, cooperative, or racing structures) independently of direct player competition.

Interaction (I). Interaction loads on axes measuring direct conflict, negotiation opportunity, player targeting, and theft/blocking. It is structurally independent of Tension because competitive games can be either high-Interaction (direct combat) or low-Interaction (parallel optimization). The separation of I and T is the empirical confirmation of the design-community intuition that "mean" and "tense" are different game properties.

Game-Architecture (G). This dimension captures production depth, engine complexity, phase count, and rulebook length — the infrastructure required to support the game's decision space. High-G games (Lacerda's catalog) require

significant setup and explanation investment. The architectural depth is orthogonal to both Tension and Elegance, confirming that complexity is not equivalent to either pressure or opacity.

Elegance (E). Elegance loads on rule parsimony, mechanical unity, and the ratio of strategic depth to rulebook length. It corresponds directly to the German-school design ideal: more decisions per rule, fewer exceptions, strong thematic integration. The negative correlation between E and G (low but consistent) reflects the design tension between depth-through-architecture and depth-through-elegance.

Range (R). Range captures the breadth of viable strategies, player-count flexibility, and scaling across skill levels. It is the least correlated with overall game weight, confirming that accessibility and replayability are determined by design choices orthogonal to both complexity and elegance. High-R games (Dominion, 7 Wonders) achieve breadth through procedural generation of the game state rather than through simplified rules.

5.2 Limitations

The normalization method assumes that every game has meaningful variation across axes (i.e., $\sigma_g > 0$ for all games). Games that score near-uniformly across all axes — perhaps because they are extremely generic designs — have near-zero denominators and are excluded from the normalized analysis. This affects approximately 2% of the corpus and does not materially affect the PCA results.

6 Conclusion

We have shown that within-game z-score normalization is the correct preprocessing step for identifying independent dimensions of game design character. Applied to the 150-game TIGRIS corpus, this normalization reduces PC1 from 81% (raw) to $\leq$ 55% and reveals 4–5 genuinely independent components explaining $\geq$ 80% of within-game normalized variance. The five components are labeled Tension, Interaction, Game-Architecture, Elegance, and Range — the TIGER dimensions formalized in TG-B1.

The methodological lesson is general: any domain where quality is a dominant latent factor in multi-attribute scores will exhibit a strong general PC that obscures within-domain variation. Within-group normalization before PCA is the standard remedy, but it has not previously been applied to game design scoring. We recommend this approach for future multi-axis evaluation studies in creative domains.

All analysis code and PCA outputs are available at [URL].

References

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